DzairVerse at HalluScoring-2026: Leak-Free QLoRA Fine-Tuning for Arabic Hallucination Detection and Truth Selection

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Abstract

Large language models answer fluently but not always truthfully, and catching a wrong answer is hardest in knowledge-sensitive domains such as Islamic question answering, where a confident but fabricated citation of Qur'an, Hadith, or jurisprudence is easy to produce and costly to miss. We study this through our participation in the HalluScoring 2026 shared task, Task 2 (Islamic Knowledge) (Alansari et al., 2026): given a question, a baseline-generated answer, and six candidate options, a system must decide whether the answer is hallucinated and then select the correct option. Our system is a single QLoRA adapter fine-tuned on Fanar-1-9B to reproduce the organizers' two-step inference, under a strictly leak-free setup, where no answer information beyond the question and generated answer enters the detection prompt, following our hypothesis that answer exposure inflates detection scores through leakage rather than judgment. On the official leaderboard our submission placed **second** of five teams, scoring 0.7582 on the blind track we targeted and 0.7881 on the assisted track. Controlled follow-up experiments support it: showing the options during detection is worth $+0.047$ Macro-F1 on both splits, and detection ability comes entirely from supervision, the frozen generator accepting its own output on all 400 test items. A 1,200-item probe, however, corrects our own initial diagnosis. The model is not blind to clean falsehoods, catching 80.5% of them; it is biased toward rejection, wrongly flagging 32.5% of verified-correct answers, and that single bias is the central limitation of the approach.

1 Introduction

Islamic question answering rewards precision and punishes plausible invention: Qur'anic references, Hadith and jurisprudence are easy to fabricate fluently and costly to get wrong. HalluScoring 2026 Task 2 (Alansari et al., 2026) turns this into a unified Arabic problem, built on the HalluTruthQA benchmark (Bouchekif et al., 2026b; Bekhouche et al., 2026) and motivated by the broader difficulty of detecting hallucination in generated text (Ji et al., 2023). Each instance gives a question, an answer from QCRI/Fanar-1-9B-Instruct and six close options (A–F); a system labels the answer, then selects the correct option, scored jointly as

$$\text{Final} = 0.5 \times \text{Macro-F1}_{\text{label}} + 0.5 \times \text{Acc}_{\text{option}}, \quad (1)$$

with the coupling rule that a wrong label makes the option wrong too. Two leaderboards are run: **A (Blind)**, where options are hidden during detection, and **B (Assisted)**, where they are visible; we targeted the harder Leaderboard A. Beyond the system we contribute the controlled experiments our original submission lacked: a leakage test, a frozen-model baseline, a large reliability probe, and evidence that the headroom left in this task lies entirely in detection.[1]

2 Background and Related Work

Data. The official Islamic splits contain 400 training, 200 development and 400 test examples in Modern Standard Arabic. Label priors differ sharply: 52.2% hallucination in training, 66.0% in development and 40.0% in test; as Section 4.4 shows, this inversion explains most of the development-to-test drop. One regularity shaped our design: in 97.5% of training and development items the generated answer copies an option verbatim, in all 259 `no_hallucination` items the copied option is the gold one, and no hallucinated item ever copies it $(0/326)$. Selection is thus trivial whenever the label is trusted. This is a property of the splits we measured, not of the task: under the same matcher the test set copies an option in only 55.8% of items.

[1] Adapter: https://huggingface.co/CH-A-Malek/haluscoring-fanar-A (access on request).

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Related work. Our approach sits at the intersection of parameter-efficient fine-tuning (Hu et al., 2022; Dettmers et al., 2023) and Arabic pretrained language models (Antoun et al., 2020; Inoue et al., 2021), now routinely fine-tuned for Arabic classification (Djemmal and Belhadef, 2025). Evaluating LLMs on Islamic knowledge is an emerging line of work: Bouchekif et al. (2025) assess LLMs on Islamic legal reasoning through inheritance law and find fluent Arabic generation coexisting with unreliable rule application, and Bouchekif et al. (2026a) formalise this into MAWARITH, a dataset and benchmark for legal inheritance reasoning. Both show that surface fluency is a poor proxy for correctness here, the gap our probe (Section 4.6) measures. The base model, Fanar-1-9B-Instruct (Fanar Team, 2025), is Arabic-centric and built on Gemma-2 (Gemma Team, 2024), the same family that produced the dataset's answers, which makes self-evaluation bias a first-order concern.

3 System

Leakage hypothesis. Letting the model see information about the answer it must judge makes detection look easy: given a reference answer, or even the options, a fine-tuned model can learn to *compare* rather than to *judge*. We therefore adopted one rule: nothing beyond the question and the generated answer ever enters a detection prompt, and the gold answer is never read. A development variant placing the reference answer in the prompt reached ≈ 0.94 but would score ≈ 0.55 under the real protocol, which never supplies it. Section 4.3 tests the milder form, option exposure.

Model. Our submission is a single QLoRA adapter (Dettmers et al., 2023) on Fanar-1-9B-Instruct, fine-tuned to imitate the organizers' two-step inference. Each instance becomes two supervised samples using the official prompts verbatim: a blind detection sample (question + answer → `{"predicted_label": ...}`, no options) and a selection sample (question + answer + options → `{"selected_option": ...}`), with completion-only loss. Algorithm 1 is the inference it reproduces. A heterogeneous leak-free pipeline (AraBERTv02 with a 4-bit LLM judge and a selector ensemble) reached only 0.772 on development and is inadmissible under the single-model rule.

Algorithm 1 Two-step leak-free inference

Require: question q, generated answer a, options O
$\ell \leftarrow \textsc{Gen}(\textsc{DetectPrompt}(q, a))$ ▷ no options shown
$label \leftarrow \textsc{Parse}(\ell)$
$s \leftarrow \textsc{Gen}(\textsc{SelectPrompt}(q, a, O))$
$opt \leftarrow \textsc{Parse}(s)$; **if** $opt \notin \{A..F\}$ **then** $opt \leftarrow A$
return $(label, opt)$

Config	Ma-F1	Lbl Acc	Opt Acc	**Comb.**
A (r16, 3ep, 2e-4)	0.808	0.830	0.810	**0.809**
C (r16, 2ep, 2e-4)	0.783	0.795	0.800	0.792
B (r32, 3ep, 1e-4)	0.795	0.805	0.780	0.787
D (r32, rsLoRA)	0.398	0.660	0.175	0.286

Table 1: QLoRA sweep on development. All runs use `paged_adamw_8bit` except C (`adamw_torch`). The four differ in more than one factor each, so single-factor effects are not identified.

Setup. Configurations are trained on the 400 training examples and scored on the 200 development examples with the organizers' verbatim harness and Eq. (1); the submitted adapter retrains the winner on all 600. Table 1 reports the four we compared (Saadi and Belhadef, 2019). Configuration A is used throughout: 4-bit NF4 QLoRA, LoRA r=16, α=32, dropout 0.05 on all attention and MLP projections, `paged_adamw_8bit`, learning rate 2e−4 cosine, warmup 0.05, 3 epochs, effective batch 16, sequence length 2048, seed 42 and `attn_implementation="eager"` (Gemma-2 softcapping), on one Kaggle T4. Figure 1 tells the same story from the optimisation side: A, B and C descend smoothly below 10^{-2}, while the rsLoRA variant D (Kalajdzievski, 2023) never leaves its ≈ 0.11 plateau and spikes at step 40, a divergence visible long before it shows in the score. Because the four differ in more than one factor each, this grid selects a configuration but does not identify single-factor effects, and we make no such claim. We build on `transformers` and `peft`; the post-hoc experiments run on two T4s in ≈5.8 GPU-hours.

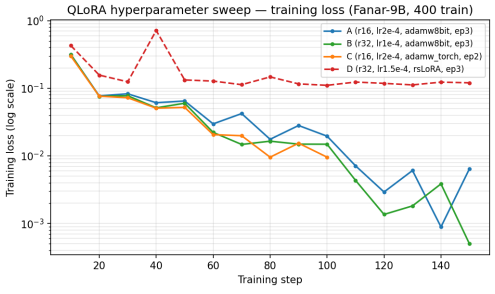


Figure 1: Training loss for the four sweep configurations (log scale).

Rank	Team	Assist. (B)	Blind (A)	B−A
1	Anupamhashtag	0.8124	0.8262	−0.014
2	**Ours (CH-A-Malek)**	**0.7881**	**0.7582**	+0.030
3	IslamicAI	0.7313	0.6541	+0.077
4	Yazid_Bounab	0.7199	0.6661	+0.054
5	Kraimia_Faiz	0.7149	0.6222	+0.093

Table 2: Official Islamic (Track 2.1) leaderboard, combined score (Eq. (1)). Only the leading system is stronger blind.

	Dev		Test	
System	Ma-F1	Comb.	Ma-F1	Comb.
Majority + random opt.	–	0.291	–	0.289
Frozen Fanar (generative)	0.254	0.412	0.375	0.544
Frozen Fanar (log-prob)	0.331	0.468	0.386	0.557
+ options shown	0.308	0.456	0.437	0.582
QLoRA, leak-free	0.811	0.793	0.754	0.784
QLoRA, options shown	**0.858**	0.814	**0.800**	0.794
Submitted adapter	–	–	0.743	0.777

Table 3: Detection and selection under identical prompts. The two QLoRA rows are trained from scratch in one session on the same 400 examples and differ only in whether the options appear in the *detection* prompt.

4 Results and Analysis

4.1 Official result

The organizers placed us **second of five** on both tracks (Table 2), at 0.7582 blind and 0.7881 assisted. The two tracks order teams differently, though: every team except the leader gains +0.03 to +0.09 from seeing the options, whereas Anupamhashtag alone scores *higher* blind, which is independent evidence for the effect measured in Section 4.3.

4.2 A frozen baseline, and self-evaluation bias

Table 3 reports frozen Fanar-1-9B under byte-identical prompts and no adapter, scored generatively (what the harness does) and by constrained log-probability comparison of the two label strings, which cannot fail to parse. The two agree, so formatting is not the story. The story is that the frozen model is *degenerate*: it answers `no_hallucination` on 400/400 test items, giving Macro-F1 0.386 against a majority-plus-random floor of 0.289. Fine-tuning adds +0.357. This also quantifies the self-evaluation bias our design invites: asked to audit its own output, the generator never once flags it, so detection ability is created by supervision, not inherited.

4.3 Leak-free versus leaky detection

We trained two adapters from scratch in the same session, on the same 400 examples, with the same seed, configuration and library versions, differing *only* in whether the options appear in the detection prompt. Showing them raises detection Macro-F1 from 0.811 to 0.858 on development (+0.047) and from 0.754 to 0.800 on test (+0.046). The effect replicates at almost the same magnitude on both splits: a model given the options learns to compare against them rather than judge the answer on its own terms. Two qualifications. The gain is confined to detection, since option accuracy *falls* slightly (−0.005 dev, −0.028 test) and combined improves by only +0.021 and +0.009. And it is learned rather than intrinsic: frozen Fanar given the same leaky prompt moves +0.051 on test but −0.023 on development. The retrained leak-free arm reaches 0.811 against 0.808 in our original sweep, reproducing the pipeline to within 0.003 on a different software stack.

4.4 Test-set diagnostics

Figure 2 gives the test confusion matrix with option accuracy in each cell. Per class the system scores P 0.656 / R 0.775 / F1 0.711 on `hallucination` (n=160) and P 0.832 / R 0.729 / F1 0.777 on `no_hallucination` (n=240): it *over-flags*, predicting `hallucination` 189 times against 160 true positives. The coupling rule punishes this twice, because in the 65 false-positive cells a wrongly rejected answer is pushed off the copy-matching path and option accuracy collapses from 0.989 (true negatives, n=175) to 0.585. Selection is right 87.3% of the time when detection is right (n=299) and 62.4% when it is not (n=101): detection errors, not selection errors, are what cost us. The prior inversion explains this. The model was selected on a split that is 66.0% hallucination and evaluated on one that is 40.0%, and a threshold calibrated to the former must over-predict the positive class on the latter; Macro-F1 duly falls from 0.808 to 0.743 while option accuracy is unchanged at 0.810.

Validity and consistency. Because option accuracy could be inflated by a default, we logged every parse: the label was unparseable in 0/400 items and the option in 2/400 (0.5%), so the reported 0.810 is not an artefact of the fallback-to-A rule. The adapter does not enforce consistency be-

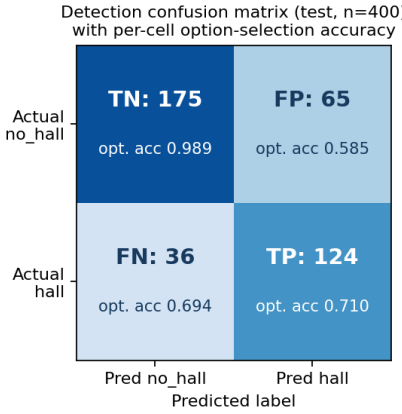


Figure 2: Test detection confusion matrix (n=400), with option-selection accuracy in each cell.

tween its steps: it predicts `no_hallucination` yet selects an option other than the copied answer in $8/135$ cases (5.9%), and predicts `hallucination` yet selects the copied answer in $30/88$ (34.1%).

4.5 Structural rules do not pay

The copy-match regularity suggests three rules: (R1) when predicting `no_hallucination`, take the copied option; (R2) when predicting `hallucination`, never take it; (R0) an answer copying no option is a hallucination. On train and development all three are exceptionless; on test all three *lose*. R1 costs 0.015 option accuracy to buy 0.003 coupled accuracy, R2 costs 0.075 (all 30 options it changes go from right to wrong), and R0 costs 0.057 Macro-F1. Each is conditioned on the *predicted* label, which is wrong for a quarter of items: R2 fires mostly on false positives, where the copied option was in fact gold, so forbidding it destroys a correct selection. With oracle detection and R1 the combined score would be 0.934: essentially all remaining headroom is in detection, none in post-processing.

4.6 What the detector actually learned

Our original analysis concluded, from ten hand-written items, that the model misses plainly false but fluent assertions. A larger probe reverses that. For each of the 400 test questions we build a *true* assertion (the gold option), a *false* one (a distractor that is neither gold nor the copied answer), and the organizers' verified free-text answer, then run blind detection on all $1{,}200$. The model labels the false assertions correctly 80.5% of the time (it is *not* blind to plain lies), but accepts only 65.5% of true assertions and 67.5% of verified-correct answers (balanced Macro-F1 0.729). The verified-answer

Outcome	Item
Missed	"What was the Battle of al-Furqān?" invents a battle (*ghazwat al-Furqān*) dated Friday 17 Ramaḍān 3 AH, passed as authentic.
Wrongly flagged	"Which grandfather raised the Prophet?" answered *'Abd al-Muṭṭalib*, correct but labelled a hallucination; selection then also failed.
Caught	A fabricated Qur'anic attribution rejected correctly, but the wrong option then selected: the coupling penalty in miniature.

Table 4: Representative test items. Arabic originals are released with the per-item error analysis.

arm is most telling: that text is free-flowing prose rather than an option string, unlike anything seen in training, yet a third of it is rejected. The failure mode is thus a bias toward rejection rather than blindness to falsehood, the same bias visible in the test precision of 0.656 and in the 65 false positives that halve option accuracy. Table 4 illustrates all three outcomes on real test items.

4.7 Reproducibility of the reported score

Our own run of the submitted adapter scored 0.777 against the organizers' 0.7582. We can bound what that gap is not: the prompts are byte-identical to those released with the adapter, and 16-bit rather than 4-bit gives 0.787, *further* from the official figure, so neither prompt wording nor quantisation explains it. What it shows is instability: identical weights and prompts differing only in precision disagree on 4.3% of labels and 8% of option choices. We treat a two-point spread as run-to-run variation and take the organizers' number as authoritative.

5 Conclusion

Our leak-free QLoRA system placed second of five teams (0.7582 blind, 0.7881 assisted). The controlled experiments reported here support the leakage hypothesis that motivated it, since option exposure buys $+0.047$ detection Macro-F1 on both splits, and show that detection ability is created by supervision rather than inherited: the frozen generator never flags its own output. They also correct our initial reading of the failure mode. The detector is not blind to fluent falsehoods, catching four in five, but is biased toward rejection, flagging roughly a third of verified-correct answers; that single bias explains the precision of 0.656, the collapse of option accuracy on false positives, and most of the development-to-test drop once the inverted prior is accounted for. Because every struc-

tural shortcut is conditioned on that unreliable label, none helps, and with oracle detection the combined score would be 0.934. The way forward is better-calibrated detection (prior-aware thresholding, training data with stylistically clean factual errors, knowledge-grounded verification) rather than better post-processing.

Limitations

Trained on 600 examples and specialised to this harness, the system is not a general Islamic QA or fact-checking model. The probe of Section 4.6 is built from option strings, stylistically in-distribution for training, though its verified-answer arm is not and points the same way. Our configuration grid does not identify single-factor effects, and the copy-match regularity holds on training and development but not on test, which we did not explain.

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